

Vojtěch Špaček

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Senior C++ / embedded engineer — real-time systems, safety-critical verification, Linux & networking | B2B contractor

Profile

- Senior **C++ and embedded** engineer working on real-time and safety-critical systems: aircraft onboard diagnostics at **Honeywell**, electron-microscopy detector software at **Thermo Fisher Scientific**, and a Linux inspection proxy and firewall at **Trusted Network Solutions**.
- Most recently sole owner of the **hardware-in-the-loop verification suite and automated bench** for a safety-certified two-stage gas furnace ignition controller, with **CAN** serial-bus and **Modbus** instrumentation throughout.
- Comfortable across the **V-model**: requirements, state-machine conformance testing, traceability and test evidence.

Availability

- **B2B contracting** — Czech IČO today, able to work with Polish counterparties (JDG if required). Remote-first; open to Kraków / Katowice on-site.
- **EU citizen (Czech Republic)** — no work permit required in Poland. English C1, Czech and Slovak native.

Core Skills

- **C++** (*Senior*, C++17/20) · **Python** (*Senior*) · **C#** (*Medior*) · **JavaScript** (*Medior*) · **Rust, Qt/QML** (*working knowledge*)
- Embedded & RTOS, ARM, real-time state machines, HAL & IPC (**COM/MIDL**, Unix sockets), FPGA-adjacent work, **gRPC**, Flatbuffers, Conan, TeamCity / GitLab CI / Jenkins, Docker
- **CAN** serial-bus diagnostics, **Modbus** I/O, **pyvisa/USBTMC/SCPI** instrument control, hardware-in-the-loop benches
- Linux, TCP/IP, **Netlink**, **nftables**, TLS 1.3 inspection, Kerberos/LDAP/OCSP, PostgreSQL, Suricata

Experience

Resideo (Ademco) - CoolPeople Contractor

Test Automation Engineer (Python, C#) (01/2026 – 09/2026): Hardware-in-the-loop verification of a safety-certified two-stage gas furnace ignition controller (**UL/CSA**, **ANSI Z21.20**, **NFPA 54**), as sole owner of the verification test suite and of the automated bench that drives it.

- Designed an **Arrange/Act/Assert** validation pipeline in **Python** with eight independent gates covering preconditions, stimulus timing, state sequence, alarms, signal history, cross-layer consistency, timer accuracy and requirement traceability.
- Built a data-driven generator expanding a **YAML** catalogue of 270+ state transitions, 38 states and 33 events into roughly 290 executable tests, removing hand-written duplication.

- Engineered the bench device layer so each instrument is owned by a single thread against a shared monotonic clock, covering `pyvisa/USBTC/SCPI` programmable AC source, `CAN` serial-bus monitoring and `Modbus` relay and probe I/O.

Thermo Fisher Scientific - Ixperta Contractor

Software Developer (C++): Electron Microscopy software development for high-resolution detector hardware operation and image acquisition.

- Implemented server-side software in `C++`, including `hardware abstraction layers (HAL)` and `inter-process communication (IPC)` using `COM` and `MIDL` on Windows.
- Developed embedded software for `FPGA` hardware, utilizing network communication protocols (`gRPC`) in `C++`, Python, and Linux environments.

Honeywell

- *Project*: Onboard Maintenance System that monitors the condition of aircraft and diagnoses issues during flight.
- Architected and implemented `DiagTest framework`, a sophisticated `state machine-based diagnostic test engine` in `C++` managing 15+ operational states for aircraft member system testing during flight and maintenance operations.

Trusted Network Solutions

- **Implemented a forward proxy** using `C++` on `Linux`, featuring `HTTP/HTTPS proxy with TLS/SSL inspection`, `OCSP integration` for certificate validation, `Kerberos authentication`, and `LDAP integration` for user/group lookup.
- **Developed and maintained ipmon**, a sophisticated network monitoring solution demonstrating deep expertise in:
- **Developed reporter_log** `C++` replacement of Perl command, implementing `PostgreSQL database integration`, `structured log parsing and processing`, and `Perl integration` for legacy compatibility with `database schema management`.

IBM — Automation Developer / IT Systems Specialist

- Automation application replacing manual ticketing and server-administration workload (**Bash**, **PowerShell**, **JavaScript**); monitored Unix/Windows production servers.

Education & Training

- **Masaryk University, Faculty of Informatics** — Computer Science coursework, *no degree awarded*.
- **Coursera / DeepLearning.AI** — Deep Learning Specialization (Andrew Ng), prompt engineering, building LLM agents. Self-directed: design patterns, Hack The Box.

Selected Open Source

- **MCP-Prompts** (97 ★, 21 forks) — Model Context Protocol server for prompt templates and LLM interactions. On npm, Docker Hub and the MCP directories; Rust port on crates.io.
- **rust-network-mgr** — Rust daemon driving `nftables` rules via `Netlink`; lock-free, epoll-based. Reimplementation of my `C++20 ipmon` daemon. On crates.io.